Stochastic Reconfiguration in Quantum Monte Carlo

Morten Hjorth-Jensen

FYS4411/9411, May 8, 2026

Content

- imaginary-time evolution and variational projection,
- tangent-space expansion of the trial state,
- derivation of the linear system $S\,\delta\boldsymbol{\alpha} = -\delta\tau\,\boldsymbol{g}$,
- interpretation of the covariance matrix S,
- relation to the natural gradient.

Setup and notation

We consider a variational wave function

$$|\Psi(\boldsymbol{\alpha})\rangle, \qquad \boldsymbol{\alpha} = (\alpha_1, \alpha_2, \ldots, \alpha_p),$$

depending on p real parameters.

The normalized state is

$$|\overline{\Psi}(\boldsymbol{\alpha})\rangle = \frac{|\Psi(\boldsymbol{\alpha})\rangle}{\sqrt{\langle\Psi(\boldsymbol{\alpha})|\Psi(\boldsymbol{\alpha})\rangle}}.$$

We wish to optimize the parameters such that the energy

$$E(\boldsymbol{\alpha}) = \frac{\langle\Psi(\boldsymbol{\alpha})|\, H \,|\Psi(\boldsymbol{\alpha})\rangle}{\langle\Psi(\boldsymbol{\alpha})|\Psi(\boldsymbol{\alpha})\rangle}$$

is minimized.

Imaginary-time evolution as a route to the ground state

The exact imaginary-time evolution of a state is

$$|\Psi(\tau)\rangle = e^{-\tau H} |\Psi(0)\rangle \, .$$

Expanding the initial state in energy eigenstates,

$$|\Psi(0)\rangle = \sum_n c_n |n\rangle \, , \qquad H |n\rangle = E_n |n\rangle \, ,$$

gives

$$|\Psi(\tau)\rangle = \sum_n c_n e^{-\tau E_n} |n\rangle \, .$$

If $c_0 \neq 0$, then for large τ,

$$|\Psi(\tau)\rangle \propto e^{-\tau E_0} |0\rangle \, ,$$

so imaginary-time propagation filters out the ground state.

Why a variational projection is needed

In Variational Monte Carlo we do not evolve an arbitrary state in the full Hilbert space. Instead, we restrict ourselves to the variational manifold

$$\mathcal{M} = \{|\Psi(\boldsymbol{\alpha})\rangle \mid \boldsymbol{\alpha} \in \mathbb{R}^p\}.$$

The exact state

$$(1 - \delta\tau H)\,|\Psi(\boldsymbol{\alpha})\rangle$$

will in general *not* belong to $\mathcal{M}$.

Idea of stochastic reconfiguration:

- apply one infinitesimal imaginary-time step,
- project the resulting state back onto the variational manifold,
- interpret the projection as a parameter update $\boldsymbol{\alpha} \to \boldsymbol{\alpha} + \delta\boldsymbol{\alpha}$.

First-order imaginary-time step

For a small time step $\delta\tau$,

$$e^{-\delta\tau H} = 1 - \delta\tau H + \mathcal{O}(\delta\tau^2).$$

Thus the exact propagated state is approximated by

$$\left|\Psi'_{\mathrm{exact}}\right\rangle \approx (1 - \delta\tau H) \left|\Psi(\boldsymbol{\alpha})\right\rangle .$$

A convenient first-order, norm-preserving form is

$$\left|\Psi'_{\mathrm{exact}}\right\rangle \approx \left[1 - \delta\tau \left(H - E(\boldsymbol{\alpha})\right)\right] \left|\Psi(\boldsymbol{\alpha})\right\rangle .$$

Subtracting $E(\boldsymbol{\alpha})$ removes the trivial component parallel to $|\Psi\rangle$.

Why subtract $E(\alpha)$?

Let

$$E = \frac{\langle\Psi|\, H \,|\Psi\rangle}{\langle\Psi|\Psi\rangle}.$$

Then

$$\langle\Psi|\,(H - E)\,|\Psi\rangle = 0.$$

Hence the vector

$$(H - E)\,|\Psi\rangle$$

is orthogonal to $|\Psi\rangle$ in the sense relevant for norm changes to first order.

Therefore,

$$[1 - \delta\tau(H - E)]\,|\Psi\rangle$$

moves the state in a direction tangent to projective Hilbert space, removing the trivial norm-rescaling component.

Variational state after a parameter update

We now expand the variational state at shifted parameters:

$$|\Psi(\boldsymbol{\alpha}+\delta\boldsymbol{\alpha})\rangle = |\Psi(\boldsymbol{\alpha})\rangle + \sum_k \delta\alpha_k \frac{\partial}{\partial\alpha_k} |\Psi(\boldsymbol{\alpha})\rangle + \mathcal{O}(\delta\alpha^2).$$

Define the tangent vectors

$$|\Psi_k\rangle \equiv \frac{\partial}{\partial\alpha_k} |\Psi(\boldsymbol{\alpha})\rangle .$$

Then

$$|\Psi(\boldsymbol{\alpha}+\delta\boldsymbol{\alpha})\rangle = |\Psi\rangle + \sum_k \delta\alpha_k |\Psi_k\rangle + \mathcal{O}(\delta\alpha^2).$$

Projection principle: minimize the distance

We compare

$$\left|\Psi'_{\rm var}\right\rangle = |\Psi\rangle + \sum_k \delta\alpha_k |\Psi_k\rangle$$

with

$$\left|\Psi'_{\rm exact}\right\rangle = [1 - \delta\tau(H-E)]\,|\Psi\rangle\,.$$

Define the mismatch vector

$$|\Delta\rangle = \left|\Psi'_{\rm var}\right\rangle - \left|\Psi'_{\rm exact}\right\rangle\,.$$

To first order,

$$|\Delta\rangle = \sum_k \delta\alpha_k |\Psi_k\rangle + \delta\tau(H-E)\,|\Psi\rangle\,.$$

The SR update is obtained by minimizing

$$\mathcal{D} = \langle\Delta|\Delta\rangle\,.$$

Expansion of the distance functional

Using

$$|\Delta\rangle = \sum_k \delta\alpha_k \, |\Psi_k\rangle + \delta\tau(H-E)\,|\Psi\rangle \,,$$

we obtain

$$\mathcal{D} = \left(\sum_i \delta\alpha_i \, \langle\Psi_i| + \delta\tau \, \langle\Psi| \, (H-E) \right) \left(\sum_j \delta\alpha_j \, |\Psi_j\rangle + \delta\tau(H-E) \, |\Psi\rangle \right) .$$

Expanding and keeping terms up to first order in $\delta\tau$,

$$\mathcal{D} = \sum_{ij} \delta\alpha_i\delta\alpha_j \, \langle\Psi_i|\Psi_j\rangle + 2\delta\tau \sum_i \delta\alpha_i \, \mathrm{Re} \, \langle\Psi_i| \, (H-E) \, |\Psi\rangle + \mathcal{O}(\delta\tau^2).$$

Stationarity condition

Differentiate $\mathcal{D}$ with respect to $\delta\alpha_i$:

$$\frac{\partial\mathcal{D}}{\partial(\delta\alpha_i)} = 2\sum_j \langle\Psi_i|\Psi_j\rangle\ \delta\alpha_j + 2\delta\tau\,\mathrm{Re}\,\langle\Psi_i|\,(H-E)\,|\Psi\rangle\,.$$

Setting this equal to zero gives

$$\sum_j \langle\Psi_i|\Psi_j\rangle\ \delta\alpha_j = -\delta\tau\,\mathrm{Re}\,\langle\Psi_i|\,(H-E)\,|\Psi\rangle\,.$$

This is the projected imaginary-time equation in the tangent basis $\{|\Psi_i\rangle\}$.

Normalized-state formulation

To remove the redundant component parallel to $|\Psi\rangle$, introduce

$$\left|\overline{\Psi}\right\rangle = \frac{|\Psi\rangle}{\sqrt{\langle\Psi|\Psi\rangle}}.$$

Define normalized tangent vectors:

$$\left|\overline{\Psi}_k\right\rangle = \frac{\partial}{\partial \alpha_k}\left|\overline{\Psi}\right\rangle.$$

Then the projected equation becomes

$$\sum_j \left\langle \overline{\Psi}_i \middle| \overline{\Psi}_j \right\rangle \delta\alpha_j = -\delta\tau \left\langle \overline{\Psi}_i \right| (H - E) \left| \overline{\Psi} \right\rangle.$$

This form leads directly to the SR metric matrix.

Logarithmic derivatives in configuration space

Let x denote a many-body configuration and write the wave function as $\Psi(x)$. Define the logarithmic derivative operators

$$O_k(x) = \frac{1}{\Psi(x)}\frac{\partial \Psi(x)}{\partial \alpha_k} = \frac{\partial \ln \Psi(x)}{\partial \alpha_k}.$$

Then

$$\frac{\partial \Psi(x)}{\partial \alpha_k} = O_k(x)\Psi(x).$$

For Monte Carlo sampling with probability

$$P(x) = \frac{|\Psi(x)|^2}{\langle \Psi | \Psi \rangle},$$

expectation values are

$$\langle A \rangle = \sum_x P(x)\, A(x)$$

(or integrals in the continuous case).

Deriving the metric matrix S_{ij}

Using normalized states, one finds

$$\left\langle \overline{\Psi}_i \middle| \overline{\Psi}_j \right\rangle = \langle O_i^* O_j \rangle - \langle O_i^* \rangle \langle O_j \rangle.$$

For real wave functions and real parameters this becomes

$$S_{ij} = \langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle.$$

Thus S_{ij} is the covariance matrix of the logarithmic derivatives.

Interpretation:

- S measures how parameter changes alter the normalized state,
- S is the metric tensor on the variational manifold,
- S is positive semidefinite.

Deriving the generalized force g_i

The right-hand side is

$$\left\langle \overline{\Psi}_i \right| (H - E) \left| \overline{\Psi} \right\rangle .$$

Define the local energy

$$E_{\rm loc}(x) = \frac{(H\Psi)(x)}{\Psi(x)}.$$

Then

$$g_i = \mathrm{Re}\left[\left\langle \overline{\Psi}_i \right| (H - E) \left| \overline{\Psi} \right\rangle \right] = \langle O_i E_{\rm loc} \rangle - \langle O_i \rangle \langle E_{\rm loc} \rangle$$

for real-valued states.
Thus g_i is the covariance between the logarithmic derivative and the local energy.

Final SR linear system

Putting everything together yields

$$\sum_j S_{ij}\, \delta\alpha_j = -\delta\tau\, g_i,$$

with

$$S_{ij} = \langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle,$$

and

$$g_i = \langle O_i E_{\mathrm{loc}} \rangle - \langle O_i \rangle \langle E_{\mathrm{loc}} \rangle.$$

In vector form:

$$S\, \delta\boldsymbol{\alpha} = -\delta\tau\, \boldsymbol{g}, \qquad \delta\boldsymbol{\alpha} = -\delta\tau\, S^{-1} \boldsymbol{g}.$$

Why this differs from ordinary gradient descent

Ordinary gradient descent updates

$$\delta\boldsymbol{\alpha}_{\mathrm{GD}} = -\eta\,\boldsymbol{g}.$$

SR instead uses

$$\delta\boldsymbol{\alpha}_{\mathrm{SR}} = -\eta\,S^{-1}\boldsymbol{g}.$$

The inverse metric S^{-1} rescales and mixes parameter directions according to the geometry of the wave-function manifold.

Consequences:

- reduced sensitivity to redundant parameterizations,
- better conditioning,
- close relation to imaginary-time evolution.

Connection to the natural gradient

The squared distance between nearby normalized states is

$$ds^2 = \sum_{ij} S_{ij}\, d\alpha_i\, d\alpha_j.$$

Thus S defines a Riemannian metric in parameter space.

Steepest descent in this metric gives

$$\delta\boldsymbol{\alpha} \propto -S^{-1}\boldsymbol{g},$$

which is exactly the **natural gradient** update.
Therefore SR is the natural-gradient method specialized to variational quantum states.

Regularization and practical solution

In practice S may be singular or ill-conditioned.
A standard fix is diagonal regularization:

$$S \;\to\; S + \lambda I, \qquad \lambda > 0.$$

Then solve

$$(S + \lambda I)\,\delta\boldsymbol{\alpha} = -\delta\tau\,\boldsymbol{g}.$$

Interpretation:

- stabilizes inversion,
- damps noisy directions,
- interpolates between SR and ordinary gradient descent for large λ.

Algorithmic summary

At each iteration:

1. Sample configurations x from $P(x) = |\Psi(x)|^2 / \langle \Psi | \Psi \rangle$.
2. Compute logarithmic derivatives $O_k(x)$.
3. Compute local energies $E_{\mathrm{loc}}(x)$.
4. Estimate
$$S_{ij} = \langle O_i O_j \rangle - \langle O_i \rangle \langle O_j \rangle,$$
$$g_i = \langle O_i E_{\mathrm{loc}} \rangle - \langle O_i \rangle \langle E_{\mathrm{loc}} \rangle.$$
5. Solve
$$(S + \lambda I)\delta\boldsymbol{\alpha} = -\delta\tau\, \boldsymbol{g}.$$
6. Update $\boldsymbol{\alpha} \leftarrow \boldsymbol{\alpha} + \delta\boldsymbol{\alpha}$.

Summary of the derivation

- Imaginary-time evolution suggests the optimal direction toward the ground state.
- The variational manifold restricts admissible states to $|\Psi(\boldsymbol{\alpha})\rangle$.
- Projecting the infinitesimal imaginary-time step onto the tangent space yields

$$S\,\delta\boldsymbol{\alpha} = -\delta\tau\,\boldsymbol{g}.$$

- S is the covariance matrix of logarithmic derivatives.
- $\boldsymbol{g}$ is the covariance of logarithmic derivatives with the local energy.
- SR is both:
  - a projected imaginary-time evolution,
  - and a natural-gradient optimization method.